5. Discuss the concept of DC offset in operational amplifiers and its effects on circuit performance.
6. How can an operational amplifier be configured as a voltage summing circuit?
7. Explain the principle of operation of a DC motor and discuss its torque-speed characteristics.
8. What factors affect the regulation and efficiency of a single-phase transformer?

Semester V

KU5DSCCSE302 JAVA TECHNOLOGIES

Semester	Course Type	Course Level	Course Code	Credits	Total Hours
5	DSC	300	KU5DSCCSE302	4	90

Learning Approach (Hours/ Week)			Marks Distribution			Duration of ESE (Hours)
Lecture	Practical/ Internship	Tutorial	CE	ESE	Total	
2	4	1	50	50	100	2(T)+3(P)*

*** ESE duration : 2 hours for theory and 3 hours for Lab**

Course Description:

By the end of this course, students will have a solid foundation in Java programming and web development concepts, equipping them with the skills needed to develop robust Java applications and web solutions.

Course Objectives:

- Fundamental Understanding of Java
- Master multithreading concepts and techniques.
- Understand AWT's abstract window toolkit.
- Understand JDBC architecture, JDBC drivers, and steps to connect to a database.

Course Outcomes:

At the end of the Course, the Student will be able to:

SL #	Course Outcomes
CO1	Understanding Java Basics and students will gain proficiency in object-oriented programming concepts
CO2	Proficiency in multithreading concepts, expertise in performing input and output operations and developing skills in GUI programming using Java AWT and Swing.
CO3	Understand the architecture of JDBC and various JDBC drivers, and learn how to manage database transactions and handle errors effectively using JDBC
CO4	Develop skills in creating, deploying, and managing Java servlets, gain proficiency in JavaServer Pages (JSP).

Mapping of COs to PSOs

	PSO1	PSO2	PSO3	PSO4	PSO5
CO1	✓	✓	✓		✓
CO2	✓	✓	✓	✓	✓
CO3	✓	✓	✓	✓	✓
CO4	✓	✓	✓	✓	✓

COURSE CONTENTS

Module 1

Features of Java, Byte code, JDK, JRE, JVM, Data Types -Type Conversion and Casting, Variables, Operators, Control Statements, Looping Statements, Arrays, Strings, Class, Objects-Declaring Objects, Access Specifier, Static, Nested and Inner Classes, this Keyword, Garbage Collection, Methods- method overloading, Constructors, Inheritance-Method Overriding, Dynamic Method Dispatch, Abstract Classes, Interface- Defining an Interface, Implementing Interfaces, Packages - Declaring a package, Importing Packages, Sub-packages, Exception Handling - Types of Exceptions, try-catch, throw, throws, and finally (20 hours)

Module 2

Thread - Life cycle of a thread, Synchronization, Thread class & Runnable interface, Multithreading. I/O streams, File streams: File Input Stream and File Output Stream, Data Input and Output Streams, Buffered Input and Output Streams Applets- Applet life cycle, working with Applets, Working with Graphics: Abstract Window Toolkit (AWT) - Components: Container, Panel, Window, Frame. AWT Controls, Listeners, Layout Managers, Event Handling - Events, Event Sources, Event Classes, Event Listener Interfaces, Adapter Classes. Swing (JFC) - Swing components (20 hours)

Module 3

Database connectivity - JDBC architecture, JDBC Drivers, Steps to connect to Database- Connectivity with MySQL, Driver Manager, Types of JDBC statements: Statement, Prepared statement, Callable statement, ResultSet - Types of ResultSet, blobs and clobs,metadata - Database Metadata, Resultset Metadata, transactions, error handling (20 hours)

Module 4

Model-View-Controller (MVC) Architecture, Java Servlets: Introduction to servlet, Servlet life cycle, Developing and Deploying Servlets, Generic and http servlets, GET, POST, HEAD and other requests, Servlet responses, error handling, security, servlet chaining, cookies, session tracking , Java Server Pages: Introduction to JSP, JSP life cycle, JSP expressions and declarations, Directives and Actions: Page directives, JSP actions and implicit objects, JSP Tag Libraries: Standard and Custom Tag Libraries, Expression Language (EL) (20 hours)

Module X :

Spring Framework: Introduction to Spring: Dependency Injection (DI) and Inversion of Control (IoC),Spring AOP (Aspect-Oriented Programming),Spring MVC :Configuring Spring MVC, Handling web requests, Hibernate: Introduction to Hibernate :Object-Relational Mapping (ORM) Hibernate architecture , Mapping in Hibernate :Mapping Java classes to database tables, HQL (Hibernate Query Language)

Core Compulsory Readings

1. Herbert Schildt, The complete reference Java2 ,11thed, Released December 2018
1. Publisher(s): McGraw-Hill ISBN: 9781260440249
2. JasonHunder& William Crawford, Java Servlet Programming, O'REILLY, 2002
3. Marty Hall, Larry Brown Core Servlets and Java server pages. Vol 1: Core Technologies. 2nd Edition.

Core Suggested Readings

1. David Flanagan, Java in a Nutshell A desktop quick Reference, 7 Edition, 2018. OReilly& Associates Inc
2. Rajkumar, Java programming, Pearson, 2013
3. Harimohan Pandey, Java Programming, Pearson, 2012
4. David Flanagan, Jim Parley, William Crawford & Kris Magnusson , Java Enterprise in a nutshell- A desktop Quick reference -O'REILLY, 2005
5. Stephen Ausbury and Scott R. Weiner, Developing Java Enterprise Applications, Wiley-2001
6. Database Programming with JDBC and Java, Reese George, Oreilly
7. Christian Bauer, Gavin King, and Gary Gregory, “Java Persistence with Hibernate” 2015
8. Craig Walls "Spring in Action" 2022
9. Christian Bauer and Gavin King, “Hibernate in Action” 2004

TEACHING LEARNING STRATEGIES

- Lecturing, Demonstration, Digital Learning, Team Work

MODE OF TRANSACTION

- Lecture, Seminar, Discussion

ASSESSMENT RUBRICS

Refer to section 7 of FYIMP- Computational Science - Scheme and Syllabus for the 4 credit courses with 2 Credit Theory + 2 Credit Practical.

Sample Questions to test Outcomes

1. Explain the difference between JDK, JRE, and JVM. How do they contribute to the execution of Java programs?
2. Discuss the role of try-catch blocks in Java exception handling. Provide an example scenario where you would use each of the following: throw, throws, and finally.
3. What is synchronization in Java multithreading? Why is it necessary? Explain the life cycle of a thread in Java.
4. Describe the steps involved in establishing database connectivity using JDBC with MySQL

KU5DSCCSE303 COMPUTER NETWORK

Semester	Course Type	Course Level	Course Code	Credits	Total Hours
5	DSC	300	KU5DSCCSE303	4	90